Cloud Native Data Engineering

Week 6. Data Governance, BI and Data Apps

Ashish Pujari | apujari@uchicago.edu

Lecture Outline

1. Data Governance
2. Business Intelligence and Dashboard Design
3. Data Visualization and Communication
4. Interactive Data Applications
5. Appendix: BI Fundamentals & Hands-On Exercises (Excel, Tableau, Metabase)

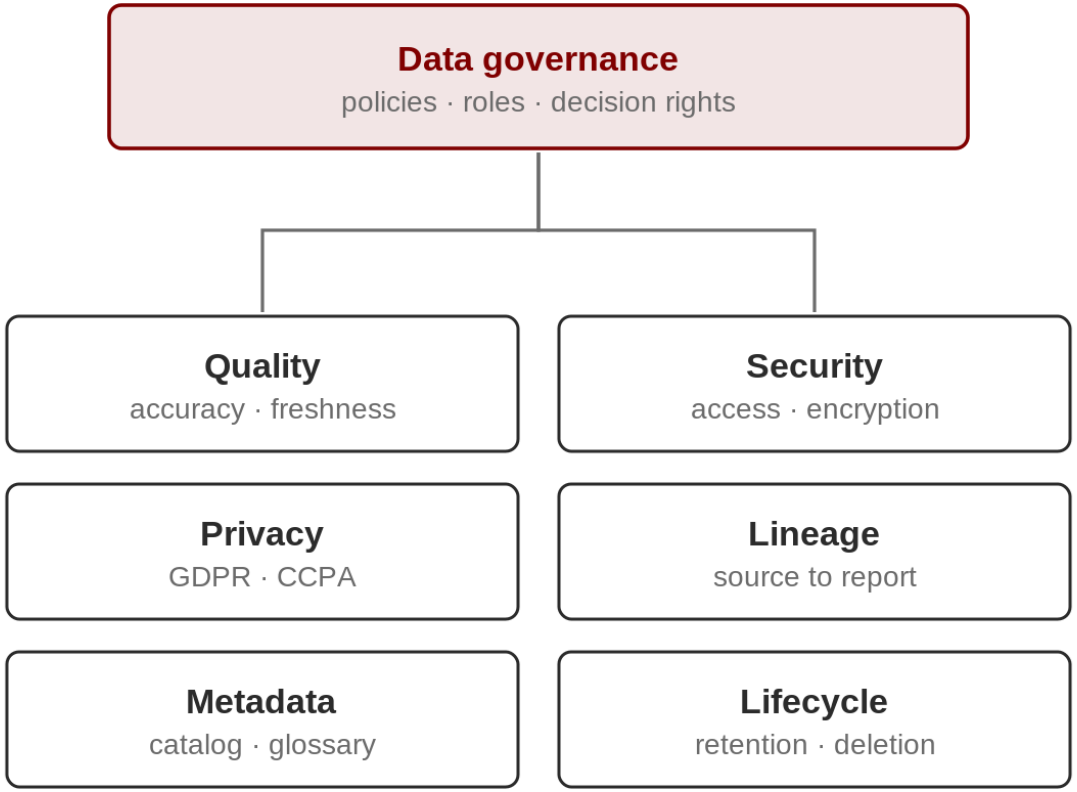


Data Governance

Module 1

Data Governance

- Authority over how data assets are managed across the organization
- Governance is an ongoing operational practice, not a one-time project
- DAMA-DMBOK: industry framework with 11 knowledge areas
- Without governance, data becomes a liability rather than an asset
- **Platforms:** Collibra, Microsoft Purview, Unity Catalog, Google Dataplex

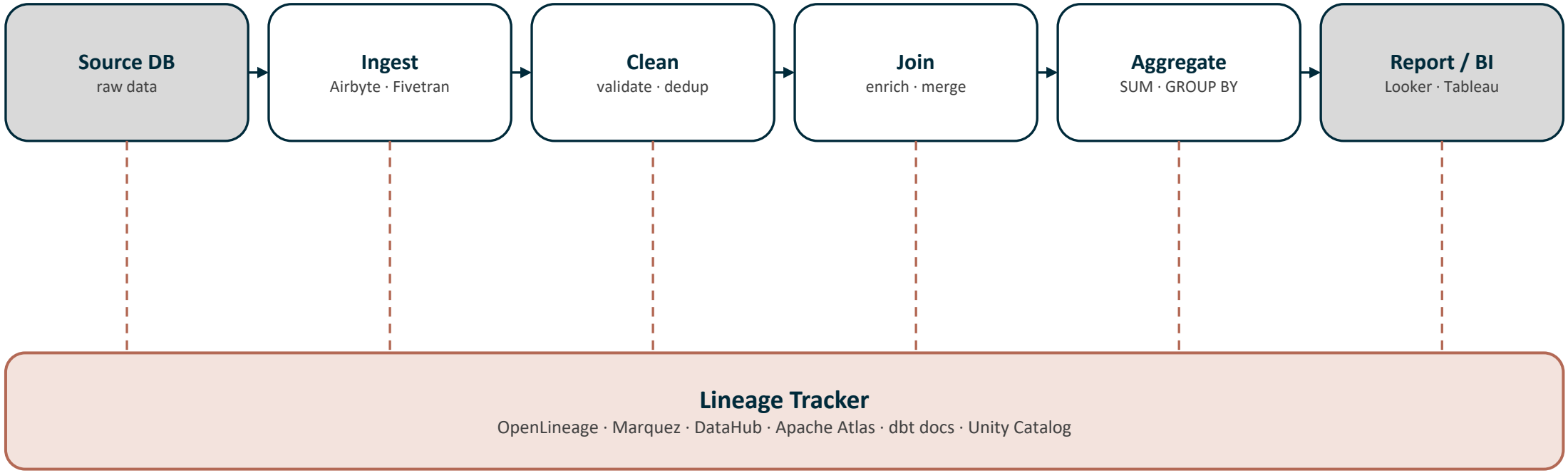


Governance sets the rules; engineering enforces them in pipelines

Data Provenance

- Provenance: the origin and history of data — where it came from and how it changed
- Captures source system, extraction timestamp, transformations applied, and author
- Essential for debugging, auditing, and regulatory compliance
- Record provenance metadata at every stage of the pipeline
- Open Lineage standard enables interoperable lineage tracking across tools
- **Example tools:** MLflow, DVC, lakeFS, W3C PROV

Data Lineage



Data lineage answers: Where did this data come from? What touched it? Who uses it?

Enables impact analysis, debugging, compliance, and trust

Source Documentation and Licensing

- Every dataset must have a documented source, access date, and license
- Open licenses: CC0, CC-BY, ODbL — check attribution and share-alike requirements
- Proprietary data: review data use agreements before using in models or reports
- Maintain a data inventory: what data exists, who owns it, what it can be used for
- Data cards / datasheets document dataset characteristics and known limitations
- **Example tools:** Hugging Face dataset cards, Croissant, Creative Commons license chooser

Privacy Regulations

Regulation	Protects	Key requirements	Engineering impact
GDPR EU, 2018	Personal data of EU residents	Access, erasure, portability; 72-hour breach notice; fines to 4% of revenue	Data residency, retention, deletion pipelines
CCPA / CPRA California, 2020	Consumer data held by larger businesses	Right to know, delete, opt out of sale or sharing	Opt-out flags, request tracking
HIPAA US, 1996	Protected health information (PHI)	Privacy and Security Rules; encryption, audit logs, breach notice	Encryption, access logs, BAAs
FERPA US, 1974	Student education records	Inspect and amend records; consent before disclosure	Role-based access, disclosure logs
LGPD Brazil, 2020	Personal data of people in Brazil	GDPR-like rights; legal basis for processing; DPO role	Consent tracking, data inventory

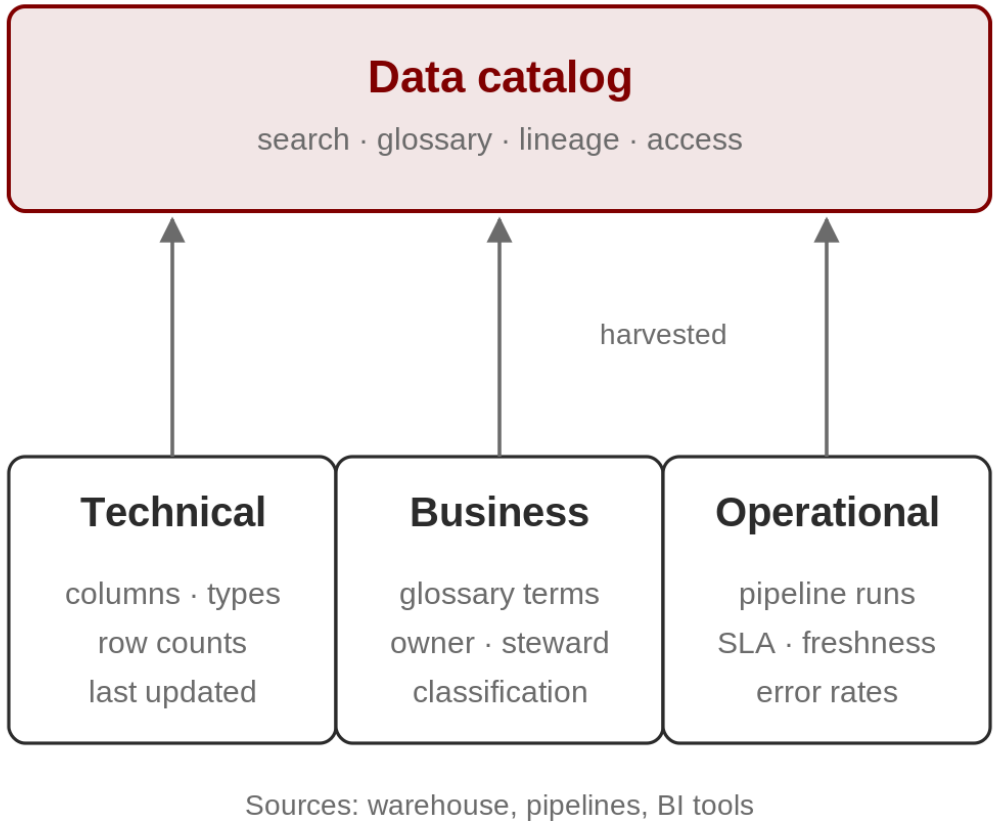
Data engineers implement these rules: storage location, retention, access logs, deletion on request
Example tools: OneTrust, BigID, Immuta, Amazon Macie

Responsible Data Use

- Fairness: check for bias in training data before building ML-driven pipelines
- Transparency: document what data is collected, stored, and for how long
- Minimization: collect only the data needed for the stated purpose
- Anonymization vs pseudonymization: different risk profiles and legal standing
- Engineers share responsibility for downstream harms from data products
- **Example tools:** Fairlearn, AIF360, Microsoft Presidio, ARX

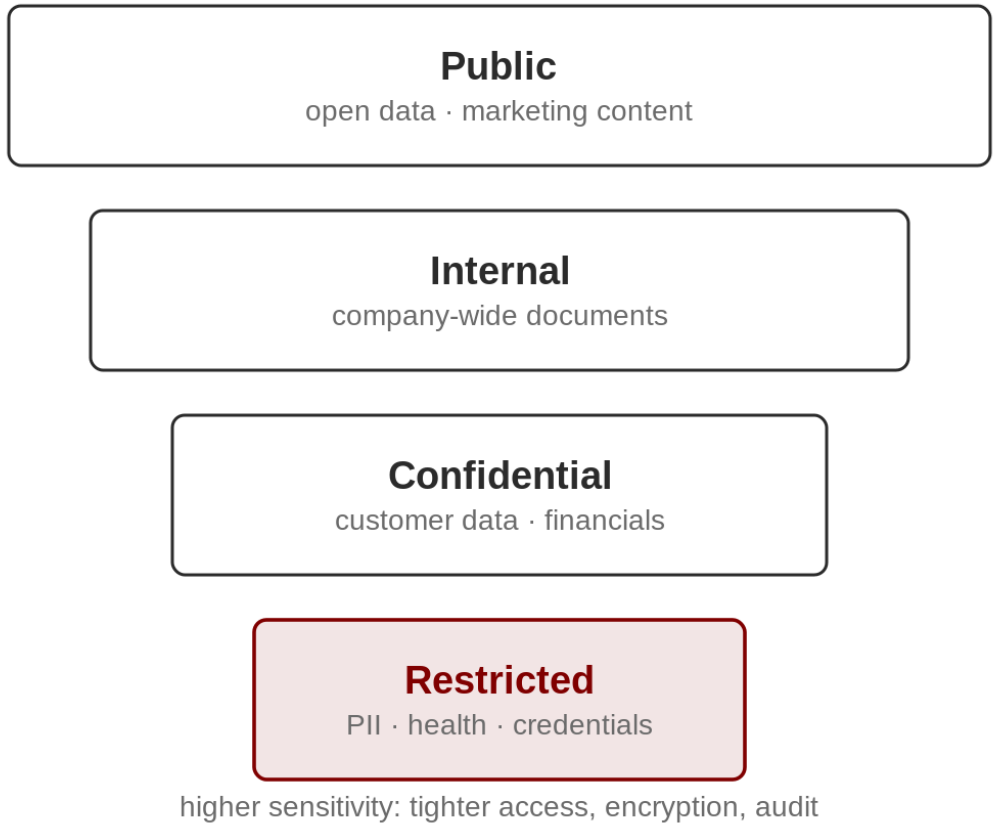
Metadata Management

- Metadata: data about data (schema, lineage, ownership, quality)
- Three kinds: technical, business, operational
- Centralize in a searchable catalog
- Example tools
  - Open source: DataHub, OpenMetadata, Amundsen
  - Cloud: AWS Glue Data Catalog, Unity Catalog, Dataplex
  - Commercial: Collibra, Alation, Atlan



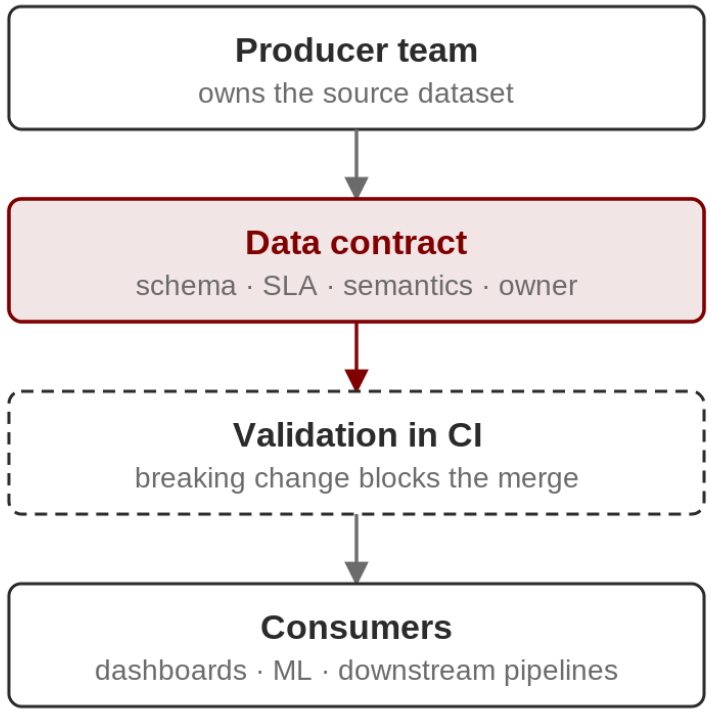
Data Classification

- Four levels: Public, Internal, Confidential, Restricted
- Classification drives access control and encryption
- Label at the column level where possible
- Review classification when data is joined or derived
- **Tools:** Amazon Macie, Microsoft Purview, Google Sensitive Data Protection, BigID



Data Contracts

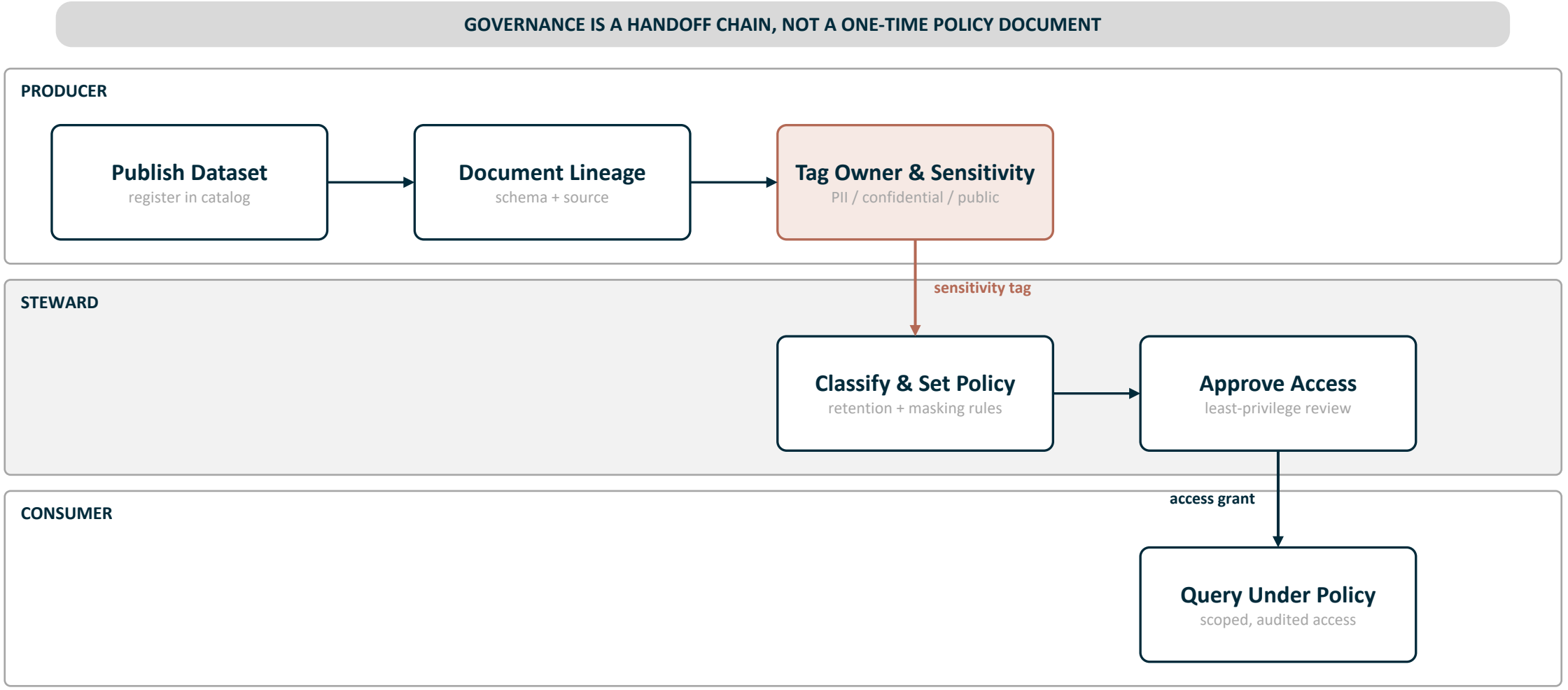
- Agreement between producer and consumer on schema, SLA, and semantics
- Specifies columns, types, freshness, and ownership
- Enforced in CI so breaking changes fail before release
- **Tools:** dbt contracts, Soda, Great Expectations, Open Data Contract Standard



Data Stewardship in Practice

- Data steward: domain expert who defines and enforces data quality for their domain
- Stewards create business glossary entries and approve changes to domain datasets
- Federated governance: each domain owns its data; central team sets standards
- Regular data quality reviews: stewards review automated quality metrics monthly
- Escalation path: data quality issues escalate to steward → data owner → CDO
- **Example tools:** Collibra, Alation, Atlan, Informatica

Governance in Practice



Governance in BI Products

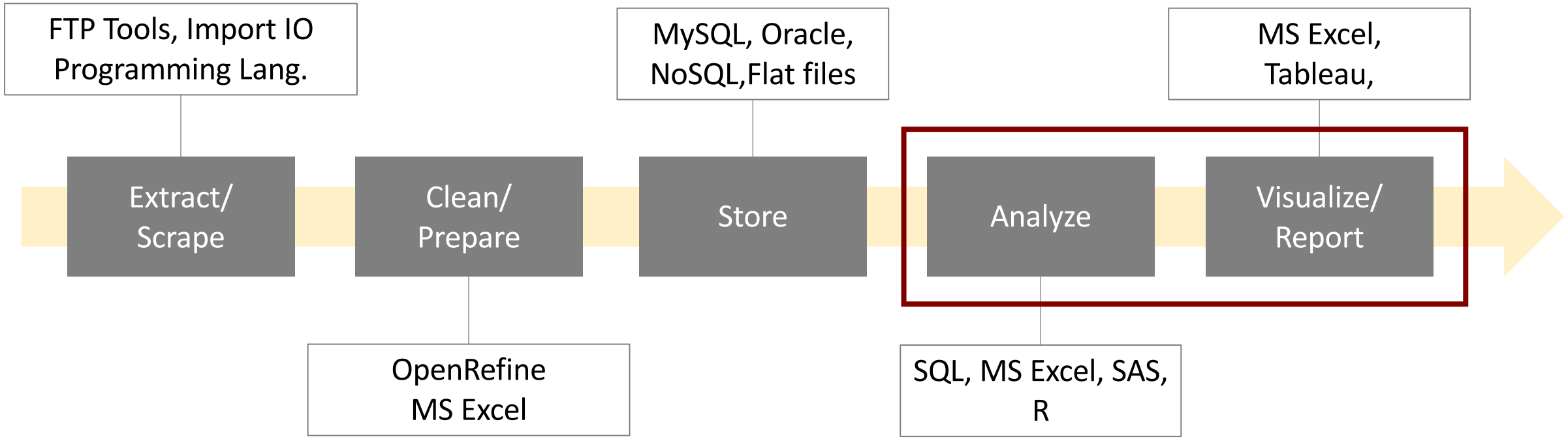
- Implement row-level security in the semantic layer, not the application layer
- Audit log all data access: who viewed which report and when
- Data freshness indicator: show users when the underlying data was last updated
- Responsible AI disclosures: document model inputs, training data, and limitations
- Review PII exposure in dashboards: mask or aggregate where appropriate
- **Example tools:** Power BI RLS, Tableau user filters, Looker access filters



Business Intelligence and Dashboard Design

Module 2

Data Analysis and Reporting



What is Business Intelligence?

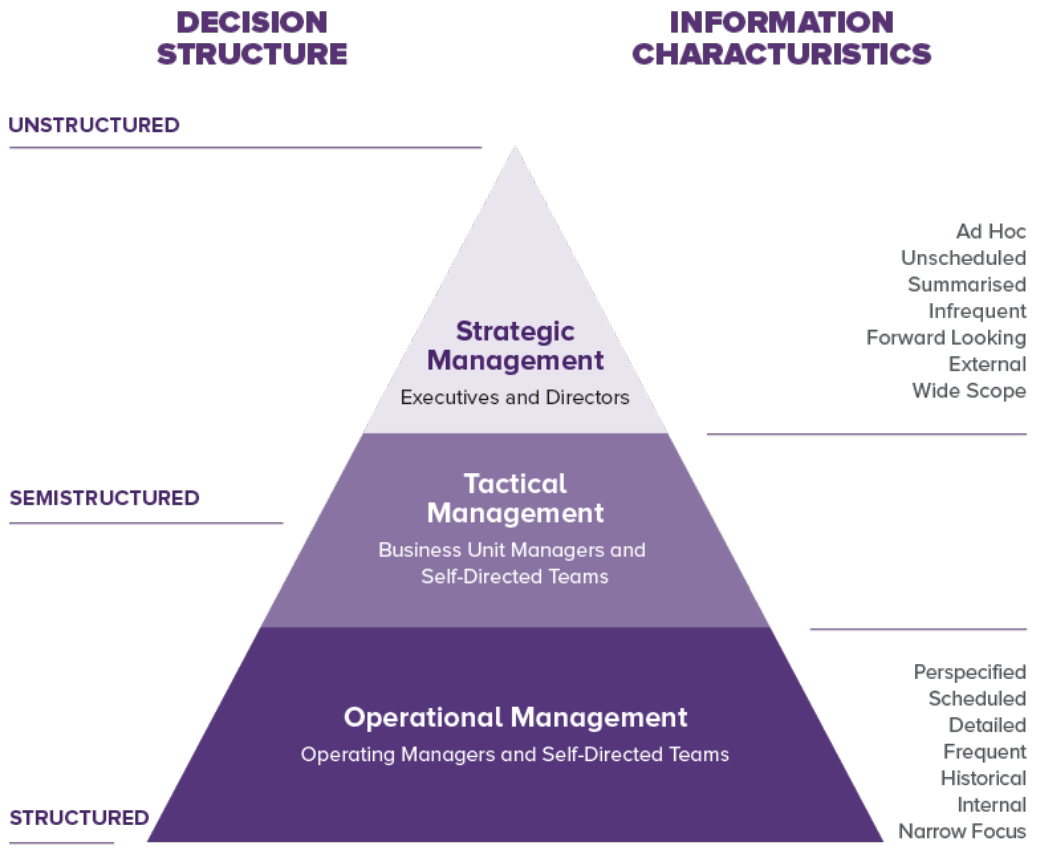
- Technologies and practices that turn data into actionable information for decisions
- Covers: reporting, dashboards, OLAP, ad-hoc query, data discovery
- Goal: right information, right people, right format, right time
- BI is not just pretty charts — it requires reliable data and well-defined metrics
- Modern BI is self-service: business users explore data without waiting for IT

BI Considerations

- What is the business problem ?
- What is the ROI in solving the business problem?
- What is the definition of success in terms of “acceptance criteria”
- Who are the audience and how are their needs different ?
- What are the dimensions for filtering and grouping data ?
- What type of metrics and KPIs can capture the business problem ?
- What are time periods of relevancy, comparison, etc. of metrics ?
- What are the SLAs for the reports/dashboards ?
- What are the security and privacy considerations ?

BI for Decision Support

- Decision Support realized as an IT system
- Provide decision support for specific business goals
- Relies on empirical information based on data
- Deliver to the right people in the right format at the right time

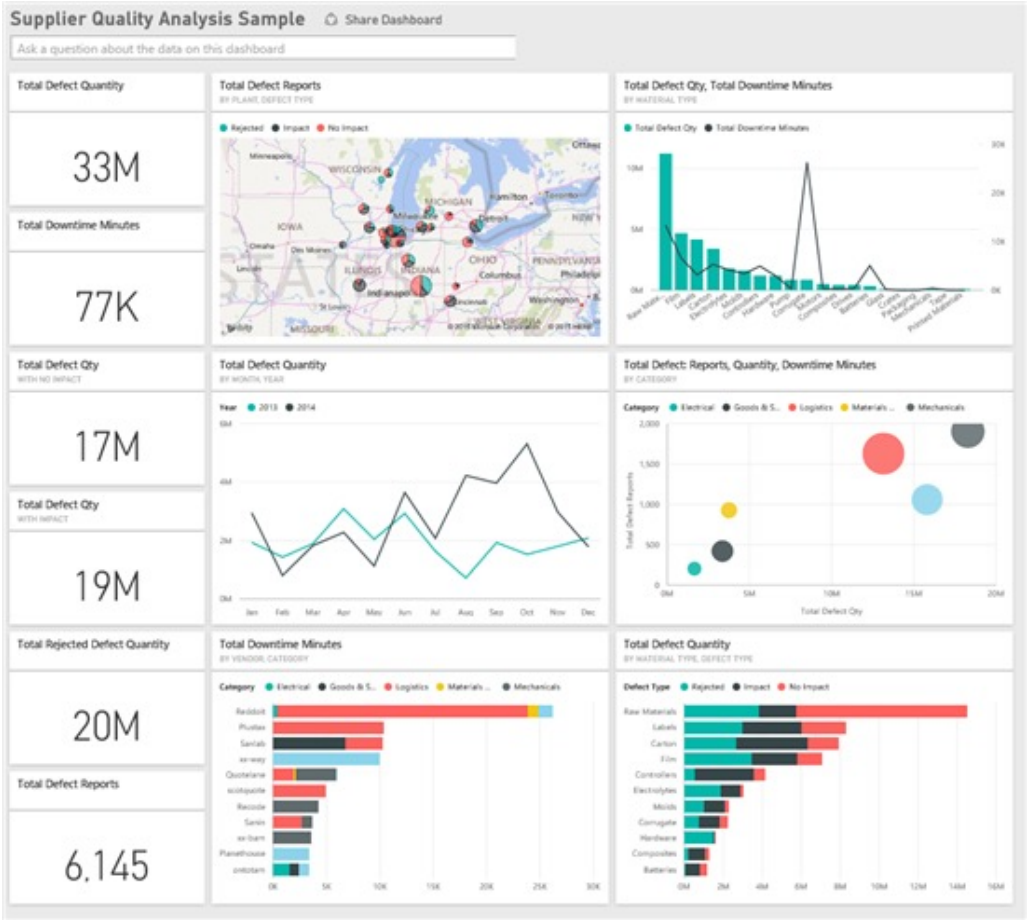


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Business Intelligence vs Standard Reporting

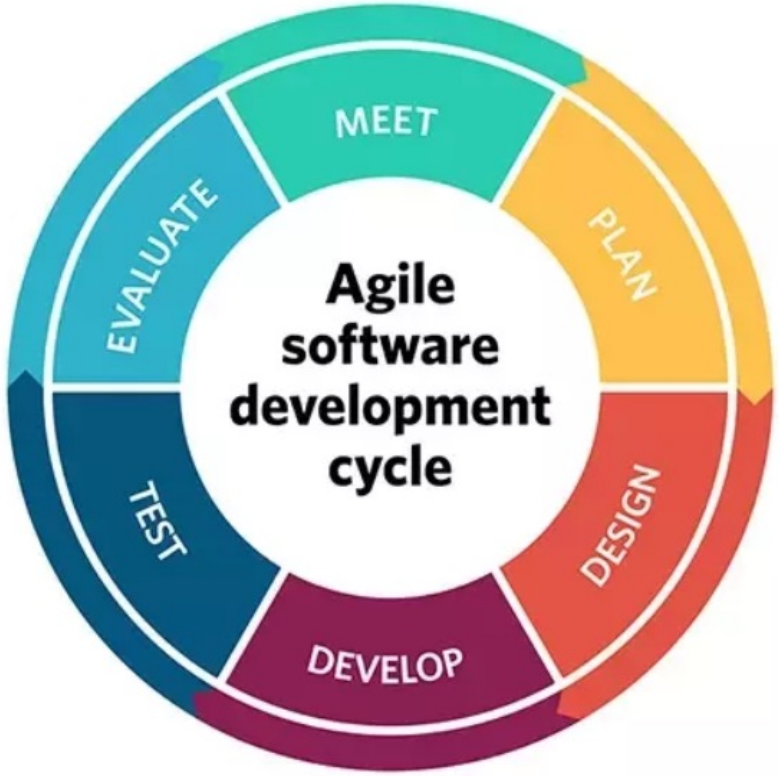
- Proactive strategies instead of retroactive reporting
- Multi-dimensional data handling
- Multi-property data selection and analysis
- Advanced visualization and export options
- Forecasting and trend analysis
- Real time or frequently updated data
- BI primary uses OLAP and dimensional models



https://powerbi.microsoft.com

BI and Data Science Project Management

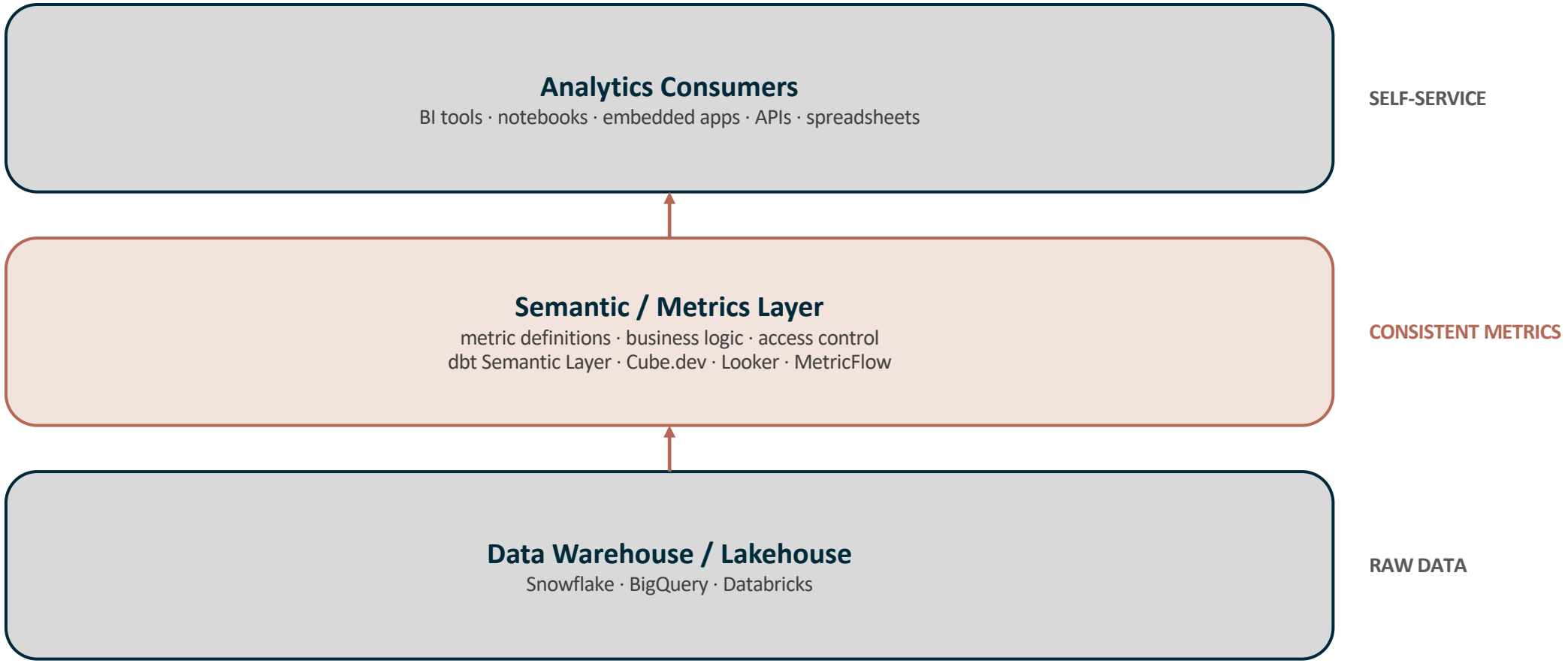
- Agile Methodologies
  - Agile Scrum
  - Agile Extreme Programming
  - Agile Kanban
- Agile Project Management Tools
  - JIRA
  - Github Projects
  - Asana
- Source Control Management
  - Github
  - Gitlab
  - Bitbucket



Self-Service Analytics

- Semantic layer: business-friendly names and metrics on top of raw SQL tables
- Certified datasets: governed, tested data products that analysts trust
- Glossaries: shared definitions of metrics prevent "different numbers" problems
- Row-level access controls ensure users only see data they are permitted to view
- Train users on data interpretation — technical accuracy ≠ business accuracy

Metrics Layer / Semantic Layer



'revenue' means the same thing everywhere — defined once, used by all tools

Business Metrics

- Quantifiable measure that is used to track and assess the status of a specific business process.
- Metrics should be employed to address key audiences surrounding a business, such as investors, customers, and different types of employees, such as executives and middle managers
- Examples
  - Marketing Metrics – Incremental Sales, Social Sentiment, SEO Traffic
  - Sales Metrics – Growth, Avg Profit Margin, Avg Purchase Value,
  - Financial Metrics – Debt to Equity, Working Capital, Current Ratio
  - HR Metrics – Cost Per Hire, Attrition Rate, Job Satisfaction/Engagement

Key Performance Indicators (KPIs)

- KPI is a measure of your performance against key business objectives.
- Every KPI should related to a specific business outcome with a performance measure.
- Smart Criteria helps evaluate the relevance of a performance indicator
  - Is your objective Specific?
  - Can you Measure progress towards that goal?
  - Is the goal realistically Attainable?
  - How Relevant is the goal to your organization?
  - What is the Time-frame for achieving this goal?

Dashboard Design Principles

- Audience first: design for the decision-maker, not the data engineer
- One dashboard = one question: resist cramming every metric onto one screen
- Hierarchy: KPI summary at top; drill-down detail below
- Reduce chart junk: remove gridlines, legends, and labels that add no information
- Color with purpose: use color to highlight outliers or encode a specific dimension

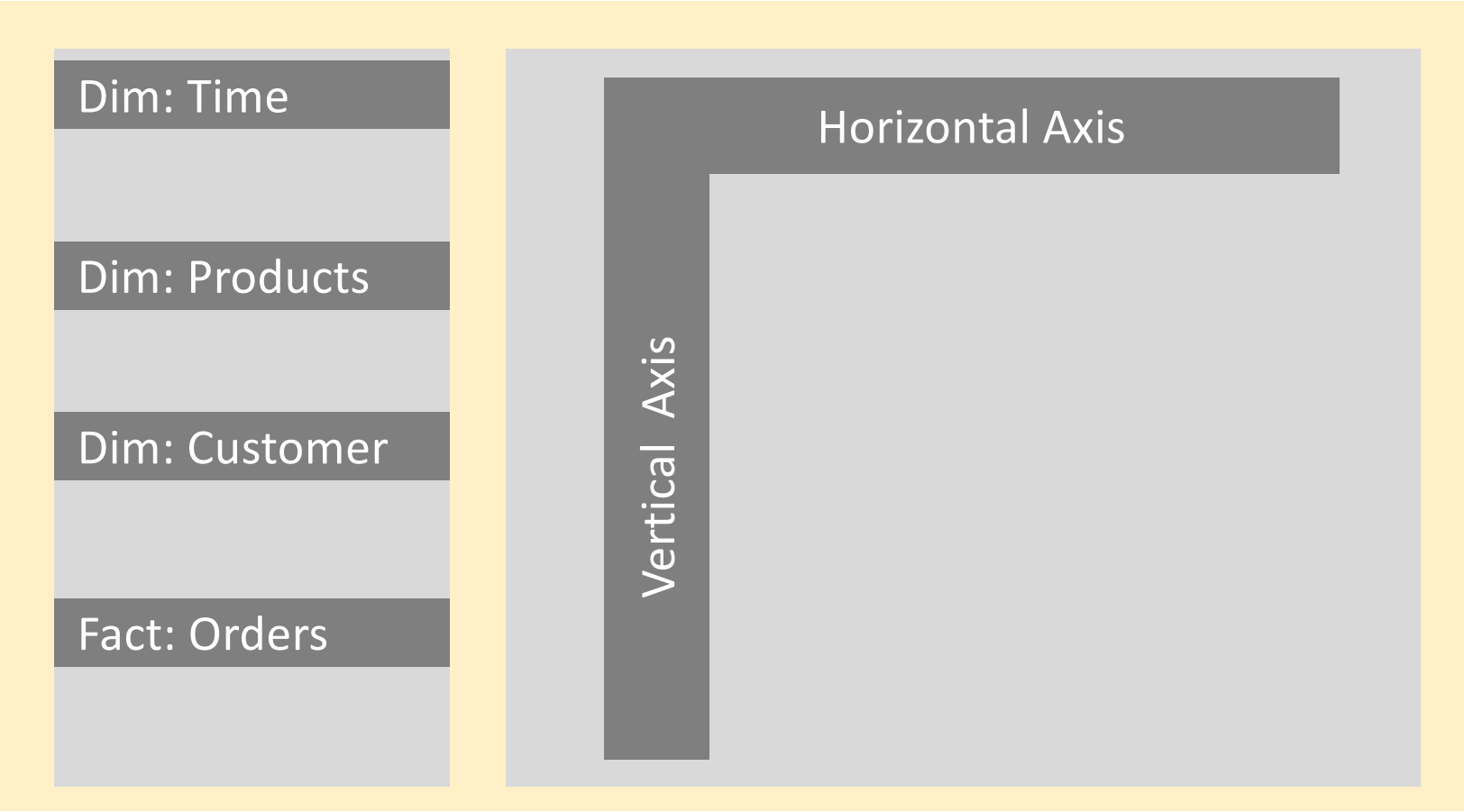
BI Platform Features

- Toggle view and drag-and-drop.
- List of native data connectors.
- Highlight and filter data.
- Share dashboards.
- Embed dashboards.
- Mobile-ready dashboards.
- Data notifications.
- Dashboard commenting.
- Create interactive dashboards.
- Create “no-code” data queries.
- Metadata management.
- Security permissions at any level.
- Translate natural language to visualizations.
- Server REST API.



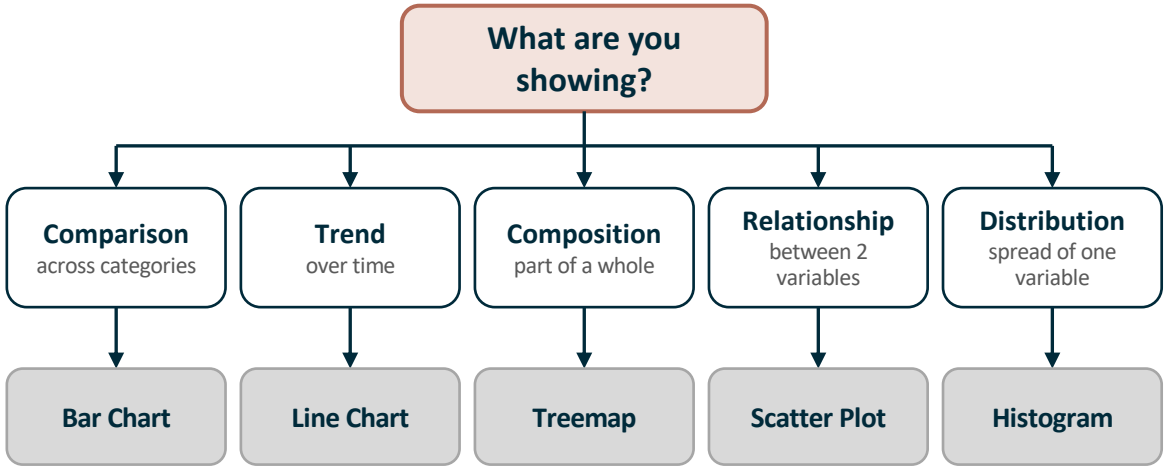
Gartner

Dashboard Anatomy



Choosing the Right Chart Type

- Time series trend → line chart; always put time on the x-axis
- Comparing categories → bar/column chart; sort by value, not alphabetically
- Part-to-whole → stacked bar or treemap; avoid pie charts for > 3 categories
- Distribution → histogram or box plot; don't summarize away the shape
- Correlation → scatter plot; add a regression line to show direction



Pick the question first; the chart type follows.

Tableau Dashboard Demo

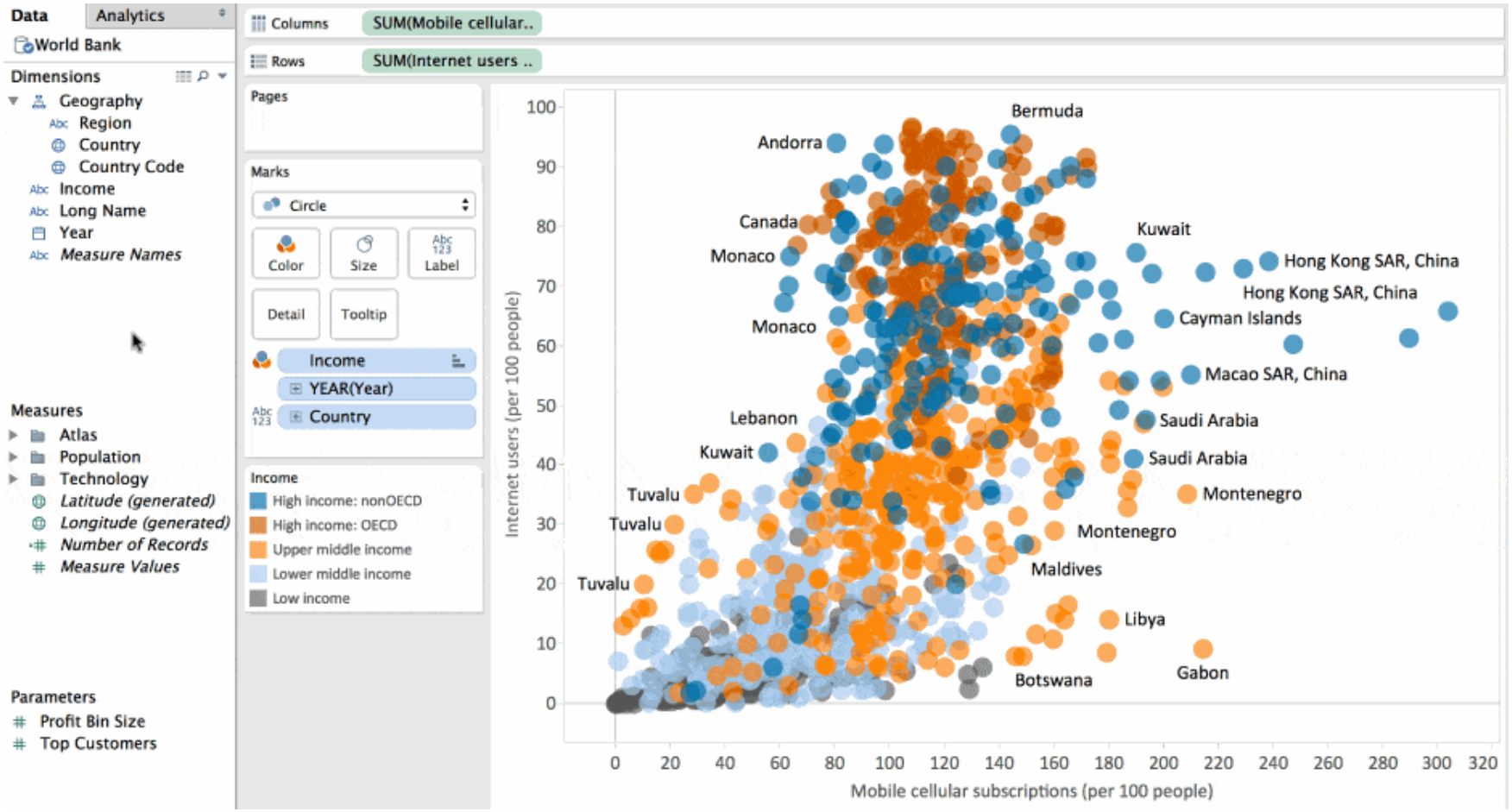
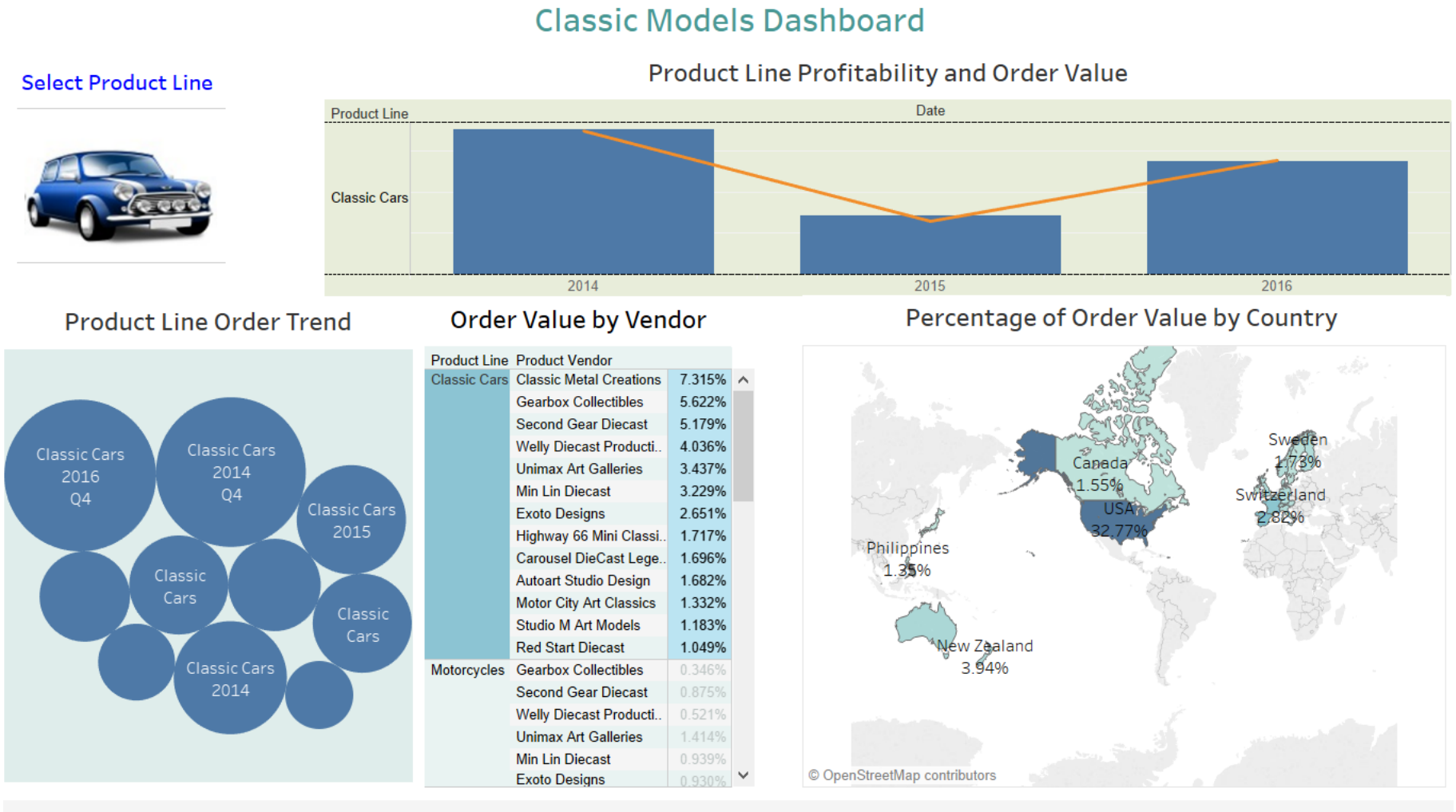
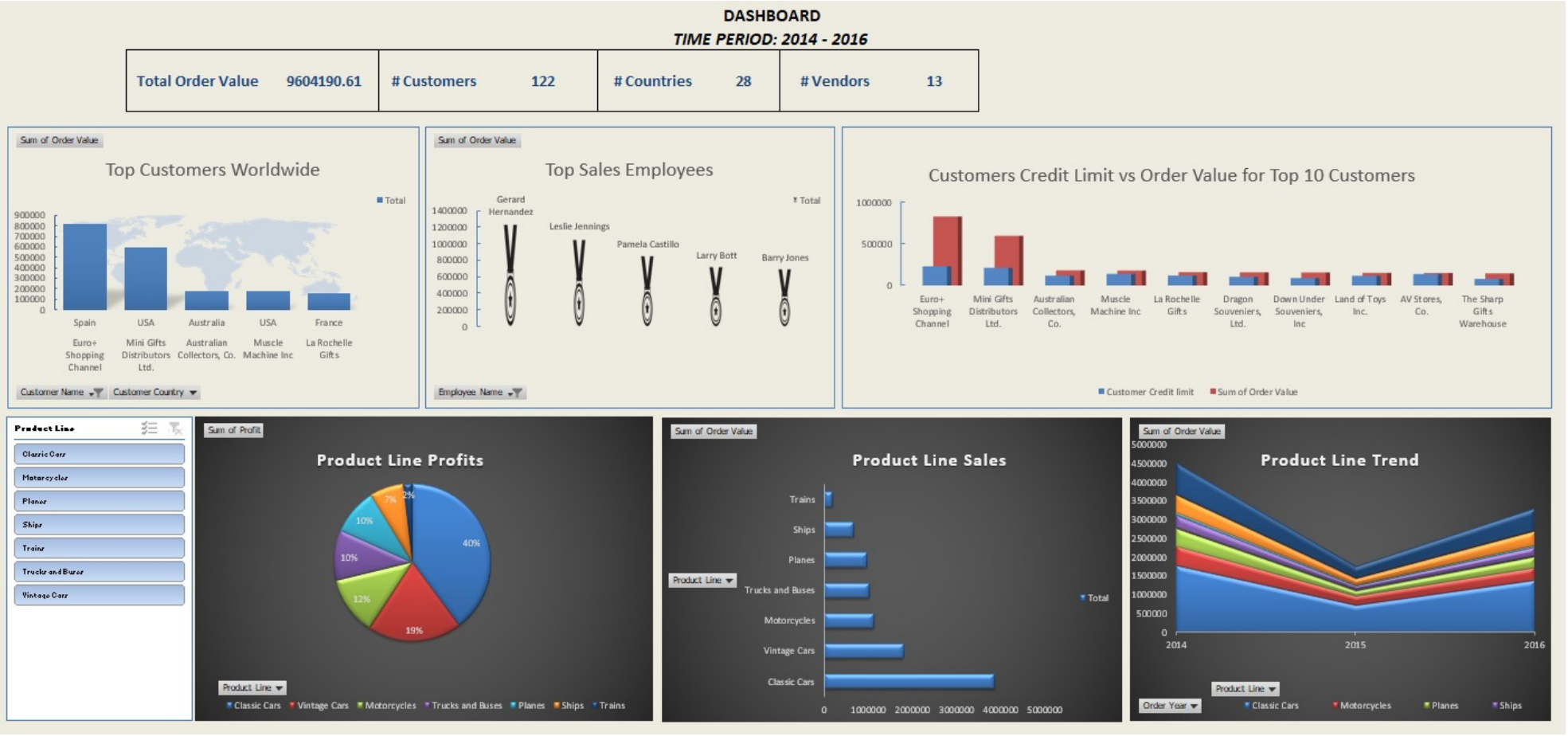


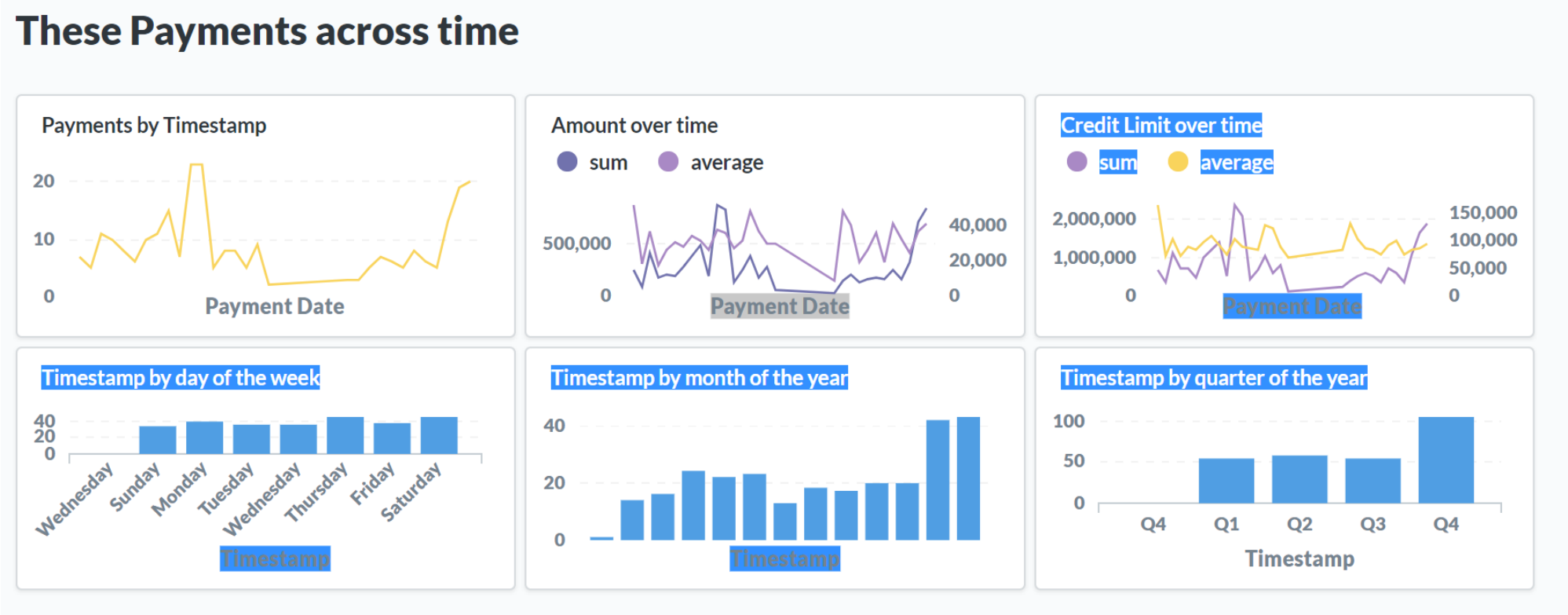
Tableau Dashboard Exercise



Excel Dashboard Exercise



Metabase Dashboard Exercise



[Metabase]



Data Visualization and Communication

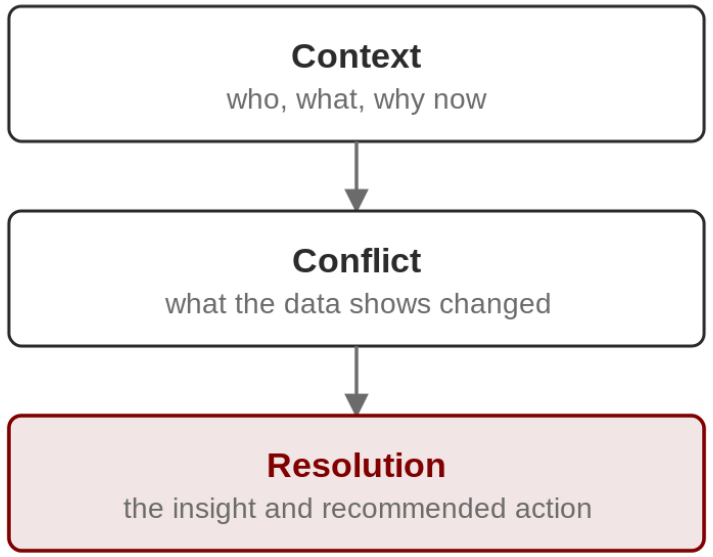
Module 3

Data Visualization Principles

- Tufte's data-ink ratio: maximize ink used for data; minimize decoration
- Pre-attentive attributes: position, color, size — processed instantly by the brain
- Gestalt principles: proximity, similarity, and enclosure guide visual grouping
- Annotation: label the most important data points directly on the chart
- Context: always include units, data source, date range, and population definition

Storytelling with Data

- Follow the narrative arc: context, conflict, resolution
- Lead with the insight, not the methodology
- Annotate inflection points on time series
- Progressive disclosure: summary first, detail on demand



Geospatial Visualization

- Choropleth map: geographic regions colored by a quantitative value
- Point map: individual events plotted by lat/lng; aggregate with hexbins for density
- Flow map: visualize movement between places (migration, logistics, transactions)
- Tools: Kepler.gl, Deck.gl, Folium (Python), Google Maps API
- Always normalize by population or area when comparing geographic regions

Temporal Data Visualization

- Line chart: continuous change over time; best for trends
- Candlestick / OHLC: financial data showing open, high, low, close per period
- Heatmap calendar: intensity of daily values over months and years
- Horizon chart: compact time series with area folded to reveal multiple series
- Annotate seasonality and key events directly on the chart

Network and Graph Visualization

- Force-directed layout: nodes repel each other; edges pull; self-organizing
- Hierarchical layout: tree structure; shows parent-child relationships clearly
- Chord diagram: shows flows between categories; good for migration or trade data
- Sankey diagram: visualizes flow quantities through stages of a process
- Tools: D3.js, Gephi, NetworkX + Matplotlib, PyVis

Dashboard Anti-Patterns

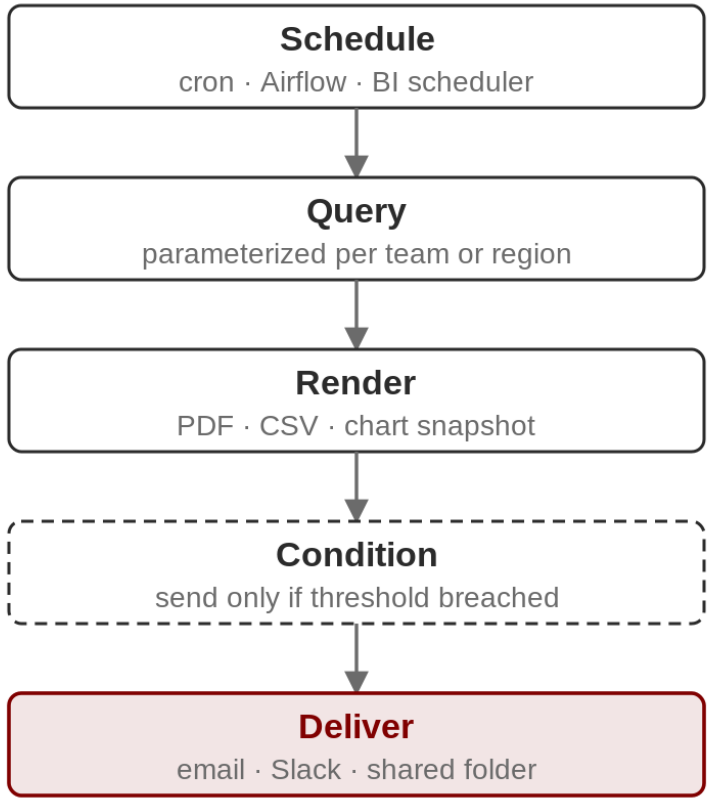
- Too many metrics: decision-makers need 3-5 KPIs, not 50 charts
- 3D charts: add no information; distort perception; never use them
- Truncated y-axis: makes small differences look enormous; always start at zero
- Rainbow color scales: perceptually non-uniform; use viridis, magma, or similar
- Missing context: a number without comparison (prior period, target) is meaningless

Dashboarding at Scale

- Dashboard performance: pre-aggregate data; use BI extract rather than live query
- Caching: cache dashboard results for N minutes; reduces warehouse load
- Staggered refresh: don't reload all tiles simultaneously — burst traffic kills performance
- Alert fatigue: limit automated alerts to < 5 per user per day; too many = all ignored
- Governance: archive stale dashboards; unused assets waste compute and confuse users

Reporting Automation

- Scheduled snapshots of key metrics on a cadence
- Conditional sends only when a threshold is breached
- Parameterize reports per region or team
- Tools: BI schedules, Python with pandas, Jinja2, smtplib



Data Product Metrics

- Adoption: daily active users, query count per dataset, dashboard views per week
- Quality: % expectations passing, freshness SLA compliance, null rate trends
- Performance: p95 query latency, slot/credit consumption, cache hit rate
- Trust: % queries using certified datasets vs uncertified ad-hoc sources
- Report these metrics to leadership; BI platform is a product that needs a roadmap

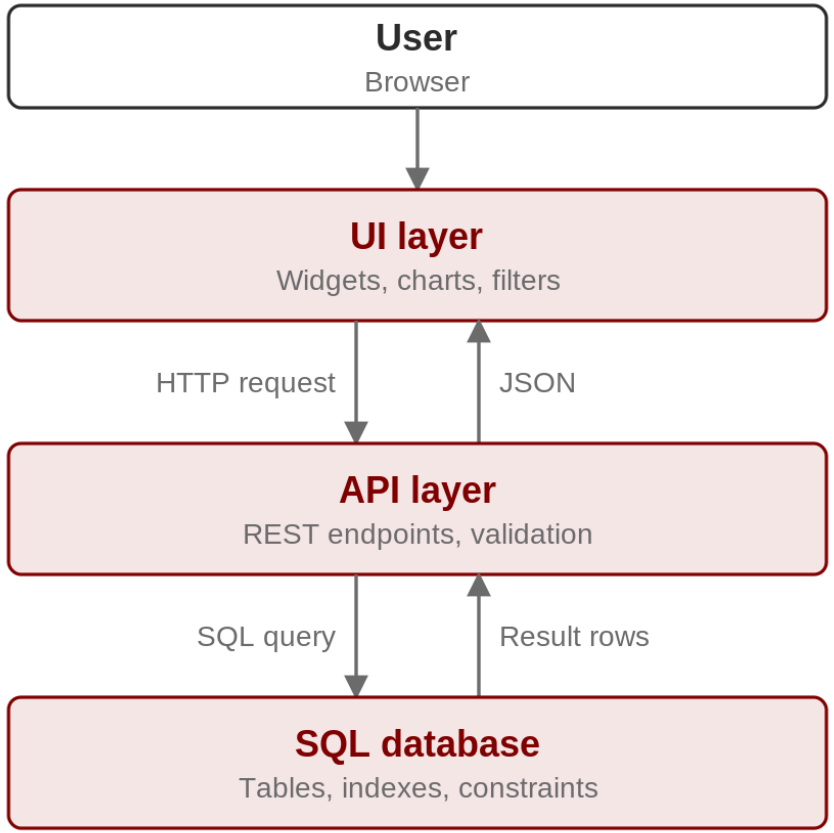


Interactive Data Applications

Module 4

Three-Tier Data Application Architecture

- UI layer: user interaction only
- API layer: business logic and validation
- SQL database: system of record
- Each tier can be swapped independently

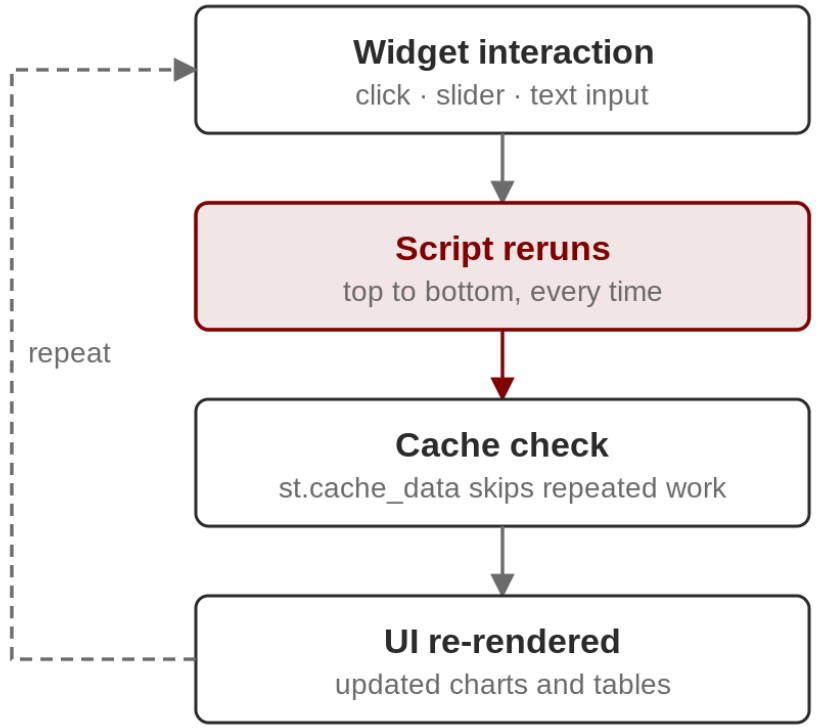


UX and Observability

- UX for data apps
  - Load progressively; use spinners for long operations
  - Filters update all dependent charts consistently
  - Explain empty states; give actionable error messages
  - Design responsively for mobile and tablet
- Observability
  - /health endpoint returns 200 when ready; wire into load balancer probes
  - Log each query: user, filters, result size, latency
  - Alert on slow queries (e.g., > 5 seconds); track feature usage

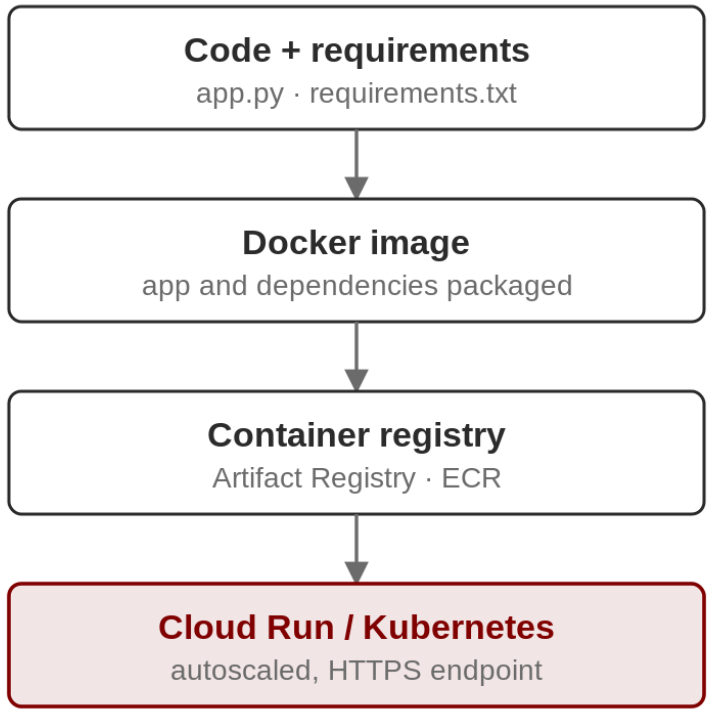
Streamlit Architecture

- Streamlit reruns the whole script on every interaction
- st.cache_data: cache function results across reruns
- st.cache_resource: share connections and models
- st.session_state: keep values between reruns



Deploying Data Applications

- Docker container: app plus dependencies
- Cloud Run: serverless; scales to zero
- Kubernetes (GKE): for complex, multi-service apps
- Deploy through CI/CD, not manually

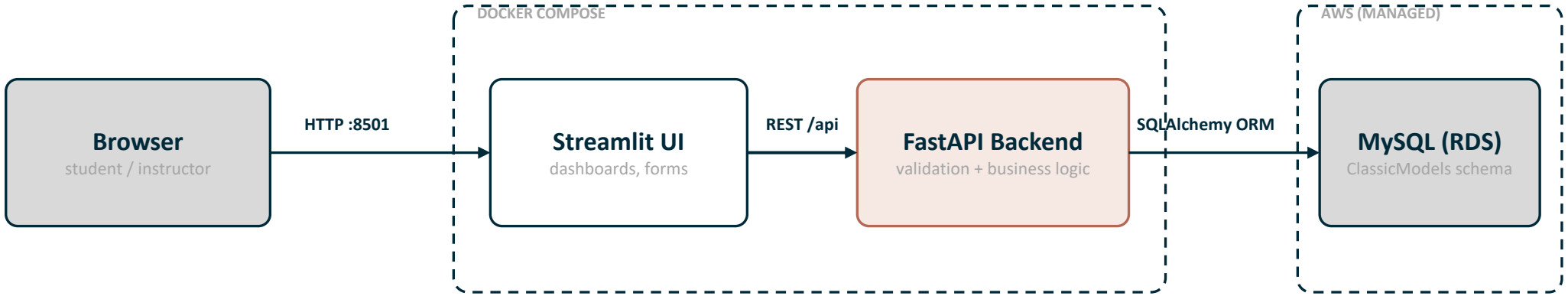


FastAPI for Data APIs

- FastAPI and Pydantic
  - Auto-generated OpenAPI/Swagger docs
  - Pydantic models validate requests and responses
- Async and dependency injection
  - async/await for non-blocking database queries
  - Shared database connections and auth context
  - Background tasks, e.g., report generation
- Course examples
  - ClassicFastAPI.py: SQLAlchemy ORM; /customers endpoint
  - ClassicFlaskAPI.py: Flask equivalent for comparison
  - SQLAlchemy: create_engine, sessionmaker, Base.query

Course App: FastAPI + SQLAlchemy + Streamlit

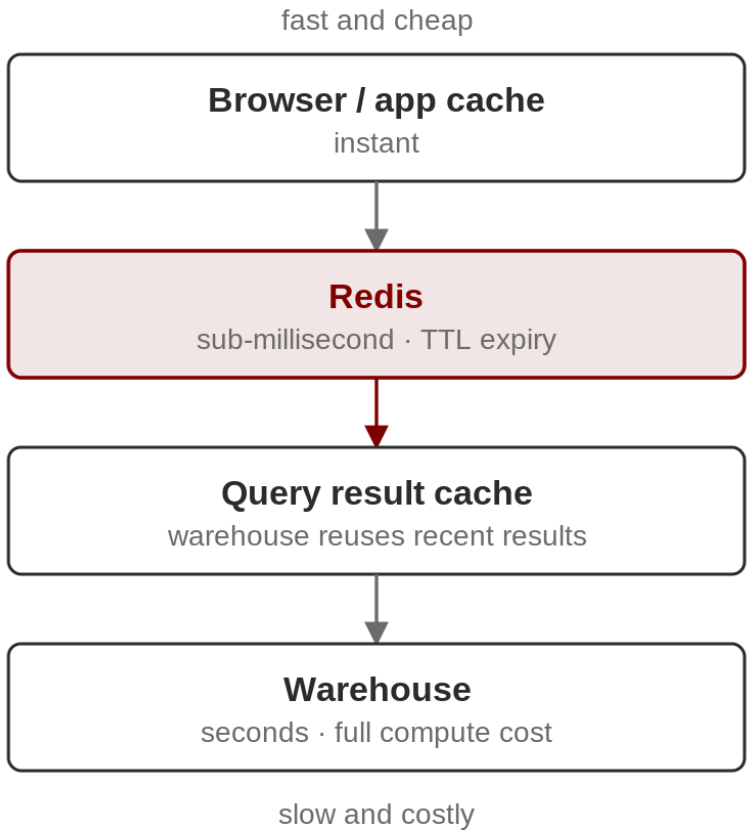
THREE-TIER APP: PRESENTATION → API → DATA, EACH SWAPPABLE INDEPENDENTLY



SAME PATTERN AS THE TEAM PROJECT'S DATA APPLICATION TIER

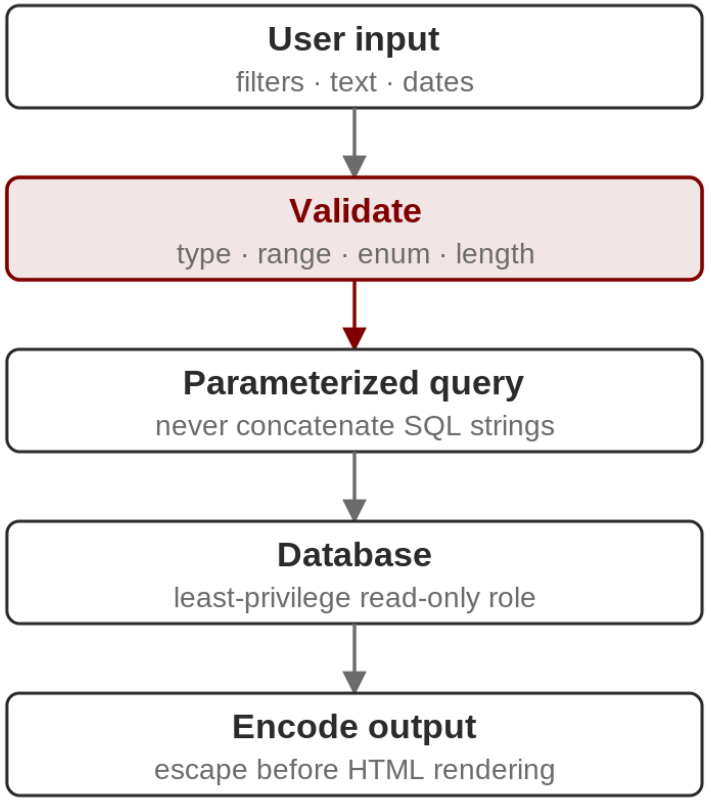
Caching Strategies

- In-memory (Redis): sub-millisecond reads with TTL
- Query result cache: reuse recent warehouse results
- Invalidate on data refresh, not only on timers
- Cache for speed and cost, but show data freshness



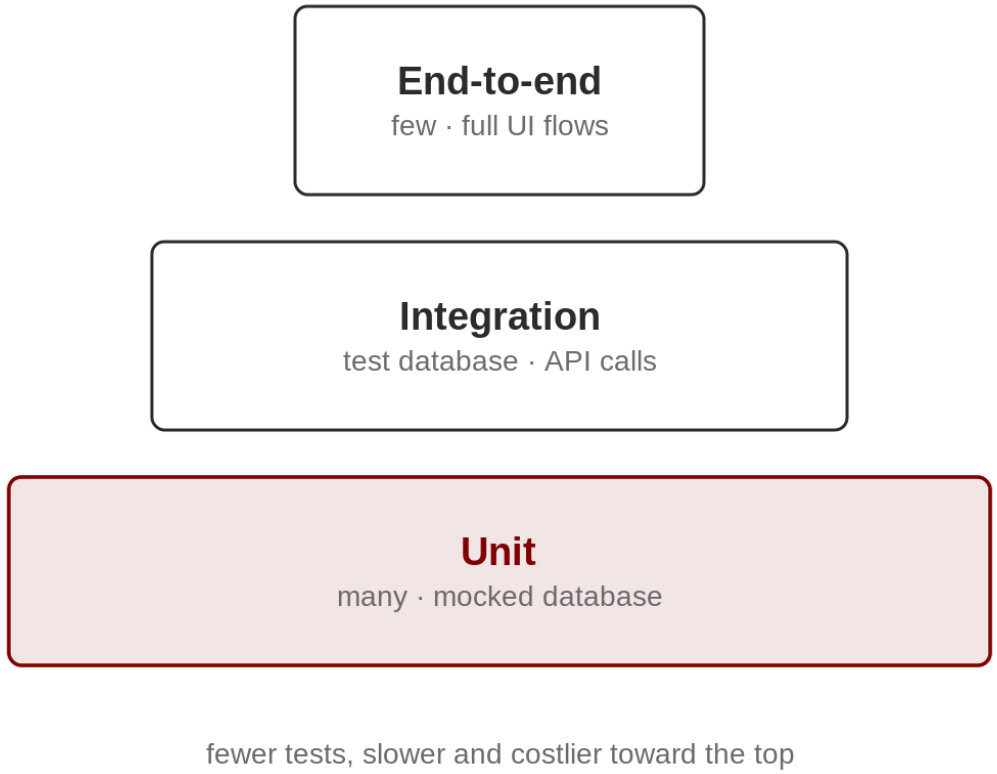
Security and Input Validation

- Authentication and authorization
  - OAuth 2.0 / OIDC via identity provider
  - Role-based access to data
- Sessions and transport
  - Short-lived signed JWTs; CSRF tokens
  - CSP, X-Frame-Options, HSTS headers
- Input validation
  - Validate first; parameterized queries only
  - Rate-limit heavy requests; encode HTML output



Testing Data Applications

- Unit tests: mock queries; test logic in isolation
- Integration tests: spin up a test database
- End-to-end: a few critical UI flows
- Run all in CI on every pull request



Summary

- Good BI starts with SMART KPIs tied to business goals.
- Design dashboards audience-first, with one question and minimal clutter.
- Match chart type to the question: trend, comparison, or distribution.
- Streamlit and Dash let Python engineers build apps without front-end work.
- BI governance needs row-level security, audit logs, and freshness checks.